

CipherRadar CBOM Scan Report

Cryptographic Bill of Materials Analysis

Target	/Users/cyberdevil/codeWorkspace/CipherRadarTestProj/
Scan Date	2026-03-18 23:15:50 +08
Duration	0.27 seconds
Files Scanned	21
Passes Run	1
Total Findings	159

Severity Breakdown

CRITICAL: 2	HIGH: 41	MEDIUM: 9	LOW: 0	INFO: 107
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Quantum Readiness Overview

Vulnerable: 20	Safe: 40	Unknown: 75	Broken: 24
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Detailed Findings

Severity	File	Line	Name	Description	Quantum Status
CRITICAL	network/tls_config.py	22	Certificate Validation Disabled	Certificate validation disabled with ssl.CERT_N...	unknown
CRITICAL	backend/auth_middlewares.js	11	none (unsigned)	JWT verify with algorithm none (unsigned) via j...	unknown
HIGH	config/app.properties	2	Hardcoded secret: database.password	Hardcoded secret found in property "database.pa...	unknown
HIGH	config/.env	6	Hardcoded secret: ENCRYPTION_KEY	Hardcoded secret found in environment variable ...	unknown
HIGH	services/data_processor.py	44	MD5	Hash function MD5 via hashlib.md5()	broken
HIGH	services/data_processor.py	39	SHA-1	Hash function SHA-1 via hashlib.sha1()	broken
HIGH	auth/password_hashing.py	8	MD5	Hash function MD5 via hashlib.md5()	broken
HIGH	auth/password_hashing.py	12	SHA-1	Hash function SHA-1 via hashlib.sha1()	broken
HIGH	services/api_client.py	40	MD5	Hash function MD5 via hashlib.md5()	broken
HIGH	certs/self_signed.pem	11	EC Private Key	PEM-encoded EC Private Key detected via regex	unknown
HIGH	certs/server.key	1	Generic Private Key (PKCS#8)	PEM-encoded Generic Private Key (PKCS#8) detect...	unknown
HIGH	backend/forge_crypto.js	50	RSA-1024	RSA key generation (bits=1024) via forge.pki.rs...	quantum-vulnerable
HIGH	network/tls_config.py	29	TLS 1.1	TLS version constraint: TLS 1.1	unknown
HIGH	network/tls_config.py	28	TLS 1.1	TLS version constraint: TLS 1.1	unknown
HIGH	backend/CryptoService.java	19	MD5	Hash function MD5 via MessageDigest.getInstance()	broken

HIGH	network/tls_config.py	15	TLS 1.0	Deprecated SSL/TLS protocol TLS 1.0	unknown
HIGH	backend/CryptoService.java	31	DES-ECB/PKCS5Padding	Cipher DES/ECB/PKCS5Padding via Cipher.getInsta...	broken
HIGH	crypto/pycryptodome_usage.py	3	SHA-1	PyCryptodome import: SHA1 from Crypto.Hash	broken
HIGH	crypto/pycryptodome_usage.py	3	MD5	PyCryptodome import: MD5 from Crypto.Hash	broken
HIGH	crypto/pycryptodome_usage.py	2	3DES	PyCryptodome import: DES3 from Crypto.Cipher	broken
HIGH	crypto/pycryptodome_usage.py	2	DES	PyCryptodome import: DES from Crypto.Cipher	broken
HIGH	crypto/encryption.py	52	RSA-1024	RSA key generation (key_size=1024) via cryptogr...	quantum-vulnerable
HIGH	backend/auth_middlewares.js	38	MD5	Hash function MD5 via crypto.createHash()	broken
HIGH	crypto/encryption.py	22	AES-ECB	Cipher algorithm AES-ECB via cryptography library	quantum-safe
HIGH	config/.env	3	Hardcoded secret: SECRET_KEY	Hardcoded secret found in environment variable ...	unknown
HIGH	config/.env	4	Hardcoded secret: JWT_SECRET	Hardcoded secret found in environment variable ...	unknown
HIGH	services/data_processor.py	45	SHA-1	Hash function SHA-1 via hashlib.sha1()	broken
HIGH	config/.env	5	Hardcoded secret: API_KEY	Hardcoded secret found in environment variable ...	unknown
HIGH	backend/crypto_utils.js	9	MD5	Hash function MD5 via crypto.createHash()	broken
HIGH	config/secrets.properties	3	Hardcoded secret: auth.api.key	Hardcoded secret found in property "auth.api.key"	unknown
HIGH	backend/crypto_utils.js	19	DES-ECB	Cipher DES-ECB via crypto.createCipheriv()	broken
HIGH	config/secrets.properties	2	Hardcoded secret: auth.jwt.secret	Hardcoded secret found in property "auth.jwt.se..."	unknown
HIGH	config/secrets.properties	4	Hardcoded secret: crypto.symmetric.key	Hardcoded secret found in property "crypto.symmm..."	unknown
HIGH	config/app.properties	7	Hardcoded secret: jwt.secret	Hardcoded secret found in property "jwt.secret"	unknown
HIGH	backend/encryption_service.ts	30	DES-ECB	Cipher DES-ECB via crypto.createCipheriv()	broken
HIGH	backend/encryption_service.ts	36	RC4	Cipher RC4 via crypto.createCipheriv()	broken
HIGH	config/app.properties	6	Hardcoded secret: ssl.keystore.password	Hardcoded secret found in property "ssl.keystor..."	unknown
HIGH	backend/encryption_service.ts	47	SHA-1	Hash function SHA-1 via crypto.createHash()	broken
HIGH	backend/forge_crypto.js	15	SHA-1	Hash function SHA-1 via forge.md.sha1.create()	broken
HIGH	config/app.properties	4	Hardcoded secret: encryption.key	Hardcoded secret found in property "encryption...."	unknown
HIGH	certs/self_signed.pem	1	RSA Private Key	PEM-encoded RSA Private Key detected via regex	unknown
HIGH	config/.env	7	Hardcoded secret: AWS_SECRET_ACCESS_	Hardcoded secret found in environment variable ...	unknown
HIGH	backend/forge_crypto.js	8	MD5	Hash function MD5 via forge.md.md5.create()	broken
MEDIUM	config/app.properties	5	TLS/SSL version: TLSv1	TLS/SSL version "TLSv1" configured in property ...	unknown
MEDIUM	config/database.yml	16	TLSv1.1	Algorithm reference "TLSv1.1" detected via regex	broken
MEDIUM	config/secrets.properties	8	TLS/SSL version: TLSv1.2	TLS/SSL version "TLSv1.2" configured in propert...	unknown
MEDIUM	config/secrets.properties	9	TLS/SSL version: TLSv1.0	TLS/SSL version "TLSv1.0" configured in propert...	unknown
MEDIUM	README.md	24	SHA-1	Algorithm reference "SHA-1" detected via regex	quantum-vulnerable
MEDIUM	README.md	25	DES	Algorithm reference "DES" detected via regex	broken
MEDIUM	README.md	23	MD5	Algorithm reference "MD5" detected via regex	broken
MEDIUM	network/tls_config.py	43	ssl.wrap_socket (deprecated)	Deprecated ssl.wrap_socket() — use ssl.SSLCon...	unknown
MEDIUM	auth/password_hashing.py	24	PBKDF2-HMAC-SHA256	Key derivation using PBKDF2-HMAC-SHA256 via has...	quantum-safe
INFO	certs/server.key	4	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown

INFO	crypto/encryption.py	15	AES-GCM	Cipher algorithm AES-GCM via cryptography library	quantum-safe
INFO	backend/webcrypto_example.js	46	ECDSA	Web Crypto generateKey with ECDSA	quantum-vulnerable
INFO	certs/ca-chain.pem	1	Certificate	PEM-encoded Certificate detected via regex	unknown
INFO	certs/ca-chain.pem	7	Certificate	PEM-encoded Certificate detected via regex	unknown
INFO	certs/ca-chain.pem	2	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/ca-chain.pem	3	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/ca-chain.pem	4	Potential key material (hex)	High-entropy hex string that may be key material	unknown
INFO	certs/ca-chain.pem	4	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/ca-chain.pem	8	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/ca-chain.pem	9	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/ca-chain.pem	10	Potential key material (hex)	High-entropy hex string that may be key material	unknown
INFO	certs/ca-chain.pem	10	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	backend/webcrypto_example.js	23	RSA-OAEP-2048	Web Crypto generateKey with RSA-OAEP-2048	quantum-vulnerable
INFO	backend/webcrypto_example.js	18	SHA-256	Web Crypto digest with SHA-256	quantum-safe
INFO	certs/self_signed.pem	6	Certificate	PEM-encoded Certificate detected via regex	unknown
INFO	certs/self_signed.pem	1	RSA	Algorithm reference "RSA" detected via regex	quantum-vulnerable
INFO	certs/self_signed.pem	4	RSA	Algorithm reference "RSA" detected via regex	quantum-vulnerable
INFO	certs/self_signed.pem	2	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/self_signed.pem	7	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/self_signed.pem	12	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/self_signed.pem	13	Potential key material (hex)	High-entropy hex string that may be key material	unknown
INFO	certs/self_signed.pem	13	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	backend/webcrypto_example.js	8	AES-GCM	Web Crypto encrypt with AES-GCM	quantum-safe
INFO	certs/server.key	2	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/server.key	3	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	backend/forge_crypto.js	55	HMAC	HMAC via forge.hmac.create()	unknown
INFO	certs/server.key	5	Potential key material (hex)	High-entropy hex string that may be key material	unknown
INFO	certs/server.key	5	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	certs/server.key	6	Potential key material (hex)	High-entropy hex string that may be key material	unknown
INFO	certs/server.key	6	Potential key material (base64)	High-entropy base64 string that may be key mate...	unknown
INFO	backend/forge_crypto.js	45	RSA-2048	RSA key generation (bits=2048) via forge.pki.rs...	quantum-vulnerable
INFO	backend/forge_crypto.js	36	AES-CBC	Cipher AES-CBC via forge.cipher.createCipher()	quantum-safe
INFO	backend/forge_crypto.js	29	SHA-512	Hash function SHA-512 via forge.md.sha512.create()	quantum-safe
INFO	backend/forge_crypto.js	22	SHA-256	Hash function SHA-256 via forge.md.sha256.create()	quantum-safe
INFO	backend/encryption_service.ts	71	Scrypt	Key derivation Scrypt via crypto.scryptSync()	unknown
INFO	backend/encryption_service.ts	66	PBKDF2-SHA256	Key derivation PBKDF2 via crypto.pbkdf2Sync()	unknown
INFO	config/app.properties	3	Algorithm reference: AES	Algorithm "AES" referenced in property "encrypt..."	unknown
INFO	backend/encryption_service.ts	61	ECDH-secp384r1	ECDH key agreement (secp384r1) via crypto.creat...	quantum-vulnerable

INFO	backend/encryption_service.ts	52	RSA-4096	RSA-4096 key generation via crypto.generateKeyP...	quantum-vulnerable
INFO	backend/encryption_service.ts	42	SHA-512	Hash function SHA-512 via crypto.createHash()	quantum-safe
INFO	backend/encryption_service.ts	19	AES-256-GCM	Cipher AES-256-GCM via crypto.createCipheriv()	quantum-safe
INFO	backend/crypto_utils.js	33	RSA-2048	RSA-2048 key generation via crypto.generateKeyP...	quantum-vulnerable
INFO	config/database.yml	8	Potential key material (hex)	High-entropy hex string that may be key material	unknown
INFO	backend/crypto_utils.js	26	AES-256-GCM	Cipher AES-256-GCM via crypto.createCipheriv()	quantum-safe
INFO	backend/crypto_utils.js	14	SHA-256	Hash function SHA-256 via crypto.createHash()	quantum-safe
INFO	backend/auth_middleware.js	26	HS384	JWT sign with algorithm HS384 via jsonwebtoken	unknown
INFO	config/secrets.properties	5	Algorithm reference: AES	Algorithm "AES" referenced in property "crypto...."	unknown
INFO	config/secrets.properties	6	Algorithm reference: RSA	Algorithm "RSA" referenced in property "signing..."	unknown
INFO	config/secrets.properties	7	Algorithm reference: SHA	Algorithm "SHA" referenced in property "hash.al..."	unknown
INFO	backend/auth_middleware.js	21	RS256	JWT sign with algorithm RS256 via jsonwebtoken	unknown
INFO	backend/auth_middleware.js	16	HS256	JWT verify with algorithm HS256 via jsonwebtoken	unknown
INFO	config/secrets.properties	10	Algorithm reference: MD5	Algorithm "MD5" referenced in property "legacy...."	unknown
INFO	backend/webcrypto_example.js	37	AES-CBC-256	Web Crypto generateKey with AES-CBC-256	quantum-safe
INFO	backend/auth_middleware.js	45	HMAC-SHA-256	HMAC with SHA-256 via crypto.createHmac()	unknown
INFO	crypto/encryption.py	31	AES-CBC	Cipher algorithm AES-CBC via cryptography library	quantum-safe
INFO	crypto/encryption.py	40	ChaCha20	Cipher algorithm ChaCha20 via cryptography library	quantum-safe
INFO	crypto/encryption.py	67	SHA-256	Hash function SHA-256 via cryptography library	quantum-safe
INFO	crypto/encryption.py	77	SHA-256	Hash function SHA-256 via cryptography library	quantum-safe
INFO	crypto/encryption.py	48	RSA-2048	RSA key generation (key_size=2048) via cryptogr...	quantum-vulnerable
INFO	backend/auth_middleware.js	31	SHA-256	Hash function SHA-256 via crypto.createHash()	quantum-safe
INFO	crypto/encryption.py	56	EC-SECP256R1	EC key generation (SECP256R1) via cryptography ...	quantum-vulnerable
INFO	crypto/encryption.py	76	HKDF	Key derivation function HKDF via cryptography l...	unknown
INFO	crypto/encryption.py	86	Scrypt	Key derivation function Scrypt via cryptography...	unknown
INFO	crypto/encryption.py	66	PBKDF2-HMAC	Key derivation function PBKDF2-HMAC via cryptog...	unknown
INFO	crypto/pycryptodome_usage.py	2	AES	PyCryptodome import: AES from Crypto.Cipher	quantum-safe
INFO	backend/CryptoService.java	41	AES key	Secret key for AES via new SecretKeySpec()	quantum-safe
INFO	backend/CryptoService.java	32	DES key	Secret key for DES via new SecretKeySpec()	broken
INFO	crypto/pycryptodome_usage.py	2	ChaCha20	PyCryptodome import: ChaCha20 from Crypto.Cipher	quantum-safe
INFO	crypto/pycryptodome_usage.py	3	SHA-256	PyCryptodome import: SHA256 from Crypto.Hash	quantum-safe
INFO	crypto/pycryptodome_usage.py	3	SHA-512	PyCryptodome import: SHA512 from Crypto.Hash	quantum-safe
INFO	backend/CryptoService.java	48	RSA	Key pair generation using RSA via KeyPairGenera...	quantum-vulnerable
INFO	backend/CryptoService.java	40	AES-GCM/NoPadding	Cipher AES/GCM/NoPadding via Cipher.getInstance()	quantum-safe
INFO	crypto/pycryptodome_usage.py	3	HMAC	PyCryptodome import: HMAC from Crypto.Hash	quantum-unknown
INFO	crypto/pycryptodome_usage.py	4	RSA	PyCryptodome import: RSA from Crypto.PublicKey	quantum-vulnerable
INFO	crypto/pycryptodome_usage.py	4	EC	PyCryptodome import: ECC from Crypto.PublicKey	quantum-vulnerable
INFO	network/tls_config.py	7	TLS	SSL/TLS context using TLS	unknown

INFO	backend/CryptoService.java	25	SHA-256	Hash function SHA-256 via MessageDigest.getInst...	quantum-safe
INFO	network/tls_config.py	20	TLS	SSL/TLS context using TLS	unknown
INFO	network/tls_config.py	27	TLS	SSL/TLS context using TLS	unknown
INFO	network/tls_config.py	34	TLS	SSL/TLS context using TLS	unknown
INFO	network/tls_config.py	8	TLS 1.3	TLS version constraint: TLS 1.3	unknown
INFO	auth/token_manager.py	15	HMAC-SHA-256	HMAC via hmac.new() with sha-256	unknown
INFO	auth/token_manager.py	11	SHA-256	Hash function SHA-256 via hashlib.sha256()	quantum-safe
INFO	network/tls_config.py	35	TLS 1.2	TLS version constraint: TLS 1.2	unknown
INFO	auth/password_hashing.py	28	HMAC-SHA-256	HMAC via hmac.new() with sha-256	unknown
INFO	auth/password_hashing.py	33	SHA-512	Hash function SHA-512 via hashlib.new()	quantum-safe
INFO	services/api_client.py	36	SHA-512	Hash function SHA-512 via hashlib.sha512()	quantum-safe
INFO	auth/password_hashing.py	18	PBKDF2-HMAC-SHA256	Key derivation using PBKDF2-HMAC-SHA256 via has...	quantum-safe
INFO	services/api_client.py	24	SHA-256	Hash function SHA-256 via cryptography library	quantum-safe
INFO	services/api_client.py	27	SHA-256	Hash function SHA-256 via cryptography library	quantum-safe
INFO	services/api_client.py	14	RSA-4096	RSA key generation (key_size=4096) via cryptogr...	quantum-vulnerable
INFO	services/api_client.py	44	TLS	SSL/TLS context using TLS	unknown
INFO	services/api_client.py	45	TLS 1.2	TLS version constraint: TLS 1.2	unknown
INFO	services/api_client.py	32	HMAC-SHA-256	HMAC via hmac.new() with sha-256	unknown
INFO	services/data_processor.py	13	SHA-384	Hash function SHA-384 via hashlib.new()	quantum-safe
INFO	services/data_processor.py	30	SHA-256	Hash function SHA-256 via hashlib.sha256()	quantum-safe
INFO	services/data_processor.py	34	SHA-256	Hash function SHA-256 via hashlib.sha256()	quantum-safe
INFO	services/data_processor.py	35	SHA-256	Hash function SHA-256 via hashlib.sha256()	quantum-safe
INFO	README.md	29	RSA	Algorithm reference "RSA" detected via regex	quantum-vulnerable
INFO	README.md	34	AES-256	Algorithm reference "AES-256" detected via regex	quantum-vulnerable
INFO	README.md	35	SHA-256	Algorithm reference "SHA-256" detected via regex	quantum-vulnerable
INFO	services/data_processor.py	46	SHA-256	Hash function SHA-256 via hashlib.sha256()	quantum-safe
INFO	services/data_processor.py	47	SHA-512	Hash function SHA-512 via hashlib.sha512()	quantum-safe
INFO	services/data_processor.py	48	SHA3-256	Hash function SHA3-256 via hashlib.sha3_256()	quantum-safe
INFO	services/data_processor.py	49	BLAKE2b	Hash function BLAKE2b via hashlib.blake2b()	quantum-safe
INFO	services/data_processor.py	23	AES-CTR	Cipher algorithm AES-CTR via cryptography library	quantum-safe

Quantum Readiness Summary

Algorithm Family	Detected Assets	Quantum Status	Recommendation
3des	3DES	broken	URGENT: Replace immediately. This algorithm is broken by classical attacks.
des	DES, DES-ECB/PKCS5Padding, DES key, D...	broken	URGENT: Replace immediately. This algorithm is broken by classical attacks.
md5	MD5, Algorithm reference: MD5	broken	URGENT: Replace immediately. This algorithm is broken by classical attacks.

rc4	RC4	broken	URGENT: Replace immediately. This algorithm is broken by classical attacks.
sha1	SHA-1	broken	URGENT: Replace immediately. This algorithm is broken by classical attacks.
tls	TLSv1.1, TLS	broken	URGENT: Replace immediately. This algorithm is broken by classical attacks.
aes	AES-256, AES-GCM/NoPadding, AES key, ...	quantum-vulnerable	Increase key size to AES-256. AES-256 provides adequate quantum resistance.
ec	EC-SECP256R1, EC	quantum-vulnerable	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
ecdh	ECDH-secp384r1	quantum-vulnerable	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
ecdsa	ECDSA	quantum-vulnerable	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
rsa	RSA, RSA-2048, RSA-4096, RSA-1024, RS...	quantum-vulnerable	Migrate to ML-KEM (FIPS 203) for key encapsulation or ML-DSA (FIPS 204) for signatures.
sha	SHA-1, SHA-256, Algorithm reference: SHA	quantum-vulnerable	Evaluate NIST PQC standards (FIPS 203/204/205) for quantum-safe replacement.
Certificate	Certificate	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
Certificate Validation Disabled	Certificate Validation Disabled	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
EC Private Key	EC Private Key	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Generic Private Key (PKCS#8)	Generic Private Key (PKCS#8)	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
Hardcoded secret: API_KEY	Hardcoded secret: API_KEY	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: AWS_SECRET_ACCESS_KEY	Hardcoded secret: AWS_SECRET_ACCESS_KEY	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: ENCRYPTION_KEY	Hardcoded secret: ENCRYPTION_KEY	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: JWT_SECRET	Hardcoded secret: JWT_SECRET	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: SECRET_KEY	Hardcoded secret: SECRET_KEY	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: auth.api.key	Hardcoded secret: auth.api.key	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: auth.jwt.secret	Hardcoded secret: auth.jwt.secret	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: crypto.symmetric.key	Hardcoded secret: crypto.symmetric.key	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: database.password	Hardcoded secret: database.password	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: encryption.key	Hardcoded secret: encryption.key	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: jwt.secret	Hardcoded secret: jwt.secret	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Hardcoded secret: ssl.keystore.password	Hardcoded secret: ssl.keystore.password	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
Potential key material (base64)	Potential key material (base64)	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
Potential key material (hex)	Potential key material (hex)	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
RSA Private Key	RSA Private Key	quantum-unknown	Migrate to ML-KEM (FIPS 203) for key encapsulation or ML-DSA (FIPS 204) for signatures.
TLS 1.0	TLS 1.0	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
TLS 1.1	TLS 1.1	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
TLS 1.2	TLS 1.2	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
TLS 1.3	TLS 1.3	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
TLS/SSL version: TLSv1	TLS/SSL version: TLSv1	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.

TLS/SSL version: TLSv1.0	TLS/SSL version: TLSv1.0	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
TLS/SSL version: TLSv1.2	TLS/SSL version: TLSv1.2	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
hkdf	HKDF	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
hmac	HMAC-SHA-256, HMAC	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
jwt	none (unsigned), HS256, RS256, HS384	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
scrypt	Scrypt	quantum-unknown	Review algorithm and assess quantum readiness based on latest NIST guidelines.
ssl.wrap_socket (deprecated)	ssl.wrap_socket (deprecated)	quantum-unknown	Migrate to ML-DSA (FIPS 204) for signatures or ML-KEM (FIPS 203) for key exchange.
blake2b	BLAKE2b	quantum-safe	No action needed. This algorithm is considered quantum-safe.
chacha20	ChaCha20	quantum-safe	No action needed. This algorithm is considered quantum-safe.
pbkdf2	PBKDF2-HMAC-SHA256, PBKDF2-SHA256, PB...	quantum-safe	No action needed. This algorithm is considered quantum-safe.
sha-256	SHA-256	quantum-safe	No action needed. This algorithm is considered quantum-safe.
sha-384	SHA-384	quantum-safe	No action needed. This algorithm is considered quantum-safe.
sha-512	SHA-512	quantum-safe	No action needed. This algorithm is considered quantum-safe.
sha3-256	SHA3-256	quantum-safe	No action needed. This algorithm is considered quantum-safe.